Background Research Document

Introduction

1. Domain

- Building maintenance requires data on installed components and equipment, including their exact location [1, Sec. 1].
- In many existing buildings this information is incomplete, obsolete, or fragmented [1, Sec. 1].
- Most existing buildings are not maintained with BIM [1].
- Creating a model of an existing building from captured data (scan-to-BIM) is costly and mainly manual [1, Sec. 2], [2].
- Camera-based reconstruction has traditionally been cheaper but less accurate than laser scanning [3, Sec. 2.2.2].
- Recent technological advances may allow camera-based reconstruction to reach higher accuracy. This is the main research gap.

2. Problem

- Heijmans maintains large buildings (e.g., offices, hospitals, educational institutions, and airports) under fixed-term contracts.
- Most of these buildings are older and have no BIM.
- At contract handover the previous maintainer provides an asset list (Excel, sometimes annotated floor plans). These records are often incomplete or outdated.
- Heijmans then performs an inventory inspection: an inspector records every asset and its condition according to NEN 2767 [4]. The inspection takes days to weeks and is monotonous.
- Current mitigations (combining data sources, AR) depend on a BIM model, which most buildings lack.

3. Goal and research questions

- Goal: a person walks through the building with a wearable camera. The captured data is converted into a 3D model of the building. An asset list can be extracted from the model.
- Requirements from stakeholders:
  - 3D model: location accuracy of 10 to 15 cm.
  - Asset list: floor and room per asset, as complete and correct as possible.
  - Data processing on premise is preferred (esp. relevant for secret projects).
- Gap that motivates SQ1: no standard method exists to assess and compare model quality [1, Sec. 4.3.4]. Benchmark datasets are lacking [3, Sec. 3.5], [2, Sec. 4].
- **Main research question:** How can a 3D model of indoor space be developed to support building inventory inspection based on visual input?
- subquestions:
  - SQ1. Which metrics evaluate methods for generating a 3D model?
  - SQ2. What methods exist and how do they compare on these metrics in literature?
  - SQ3. How do the selected methods perform in experiments and which factors affect the results?

    - (possibly later) SQ4. How can assets of one type and their condition be identified in the 3D model?

Background/Related work

1. Concepts and definitions

- Building maintenance: asset, inventory inspection, BIM (IFC), Level of Detail (LoD), COBie (optional: what an asset list should contain).
- Localisation: camera pose, SLAM, drift, loop closure, sensor fusion (e.g., camera and IMU).
- Geometry reconstruction: depth, scale ambiguity of a single camera, 3D representations (point cloud, mesh, Gaussian splats).
- Semantic identification: semantic segmentation, object detection, semantic mapping.
- Building information modelling: scan-to-BIM, as-built (as-is) model.
- 3D world models:
    - (a) a measured 3D model of the building (the goal of this project).
    - (b) learned or generative world models, which can be used as a means to reach (a).
    - Measured vs. generated content: generated parts may look plausible without being correct, which matters for the accuracy requirement.

2. Reconstruction pipeline

- Capture, camera poses and geometry, 3D representation, semantics, asset list or BIM.

3. Method families per pipeline stage

- Photogrammetry: structure from motion (SfM) and multi-view stereo (MVS).
- Robotics: simultaneous localisation and mapping (SLAM) and semantic mapping.
- Neural and learning-based methods: Gaussian splatting, feed-forward reconstruction models.
- Short history: from photogrammetry and laser scanning to SLAM to learning-based methods.

4. Robotics assumptions vs. building inspection

- Typical assumptions of robotics methods (e.g., calibrated sensors on a robot, static scenes).
- Conditions in this project: a person with a wearable camera, long corridors, glass, repetitive rooms.
- Which assumptions remain valid and which do not.

5. Available hardware for testing

Methods section of the final paper.

- Capture devices (specifications from datasheets: TODO):

Device	Type	Provided by
Samsung Galaxy S25	Smartphone	personal device
Ray-Ban Meta glasses Gen 2 (Wayfarer)	Smart glasses	Heijmans
Insta360 X6	360° camera	Heijmans

Device	Type	Provided by
DJI Osmo Pocket 3	Handheld gimbal camera	ISSD
GoPro Fusion	360° camera	ISSD
Ricoh Theta S	360° camera	ISSD

- Computing hardware:

Component	Specification
Operating system	Ubuntu 24.04.4 LTS
CPU	AMD Ryzen 9 7950X, 16 cores, 32 threads
RAM	Approximately 128 GB
GPUs	Two NVIDIA RTX 4090, approximately 24 GB VRAM each
Free storage	Approximately 42 GB on the project filesystem when checked

- Note: GPU memory does not combine across cards, so experiments should fit on one GPU.

6. Research gap

- Image-based reconstruction using recent advancements in technology has not been evaluated against the requirements of inventory inspection.
- No standard method exists to assess and compare model quality [1, Sec. 4.3.4]. Benchmark datasets are lacking [3, Sec. 3.5].

Relevant results from other papers

- Reported accuracy and completeness of each method family on indoor scenes, compared with the 10 to 15 cm requirement (input for SQ2).
- Factors known to cause failures (e.g., glass, low-texture surfaces, occlusion) (input for SQ3 test conditions).
- (optional) Results of object detection in 3D models (input for SQ4).

Inspiring methodologies from other papers

- Evaluation metrics and protocols used for 3D reconstruction (input for SQ1). Criteria: geometric accuracy, reconstruction completeness, robustness, computational performance, and suitability for downstream asset modelling.
- Ground truth: how reference data is obtained (e.g., a laser scan as reference).
- Benchmark datasets for indoor reconstruction.

Research process

This research uses the following ICT research methods from the DOT framework [5]:

Method	DOT strategy	Activity
Expert interview	Library	Heijmans stakeholders were interviewed about the maintenance process and its problems. An interview was conducted also with a maintenance expert. Fontys ICT Game Design and Video Production experts were interviewed about 3D reconstruction of physical spaces used for gaming and video production.
Problem analysis	Field	The interview results were documented in a problem analysis, which was discussed with and approved by the stakeholders.
Literature study	Library	Review papers on BIM for existing buildings, scan-to-BIM, and condition assessment were studied to describe the domain and the research gap.
Available product analysis	Library	Existing reconstruction software and commercial capture products were reviewed for their output, licence, and hardware requirements.
Benchmark test	Showroom	[TODO] Selected methods are compared on the same captured data using the metrics from SQ1.
Decomposition	Workshop	[TODO] The reconstruction process was decomposed into pipeline stages (capture, localisation, geometry, semantics, BIM) to compare methods per stage.
Prototyping	Workshop	[TODO] A prototype was built to investigate what different algorithms can do using Samsung Galaxy S25 video.
Requirements prioritization	Workshop	[TODO]

Conclusion

- What is known: summary of the method families and their reported performance.
- What is missing: the research gap.
- how each SQ addresses the gap.

References

[1] R. Volk, J. Stengel, and F. Schultmann, "Building information modeling (BIM) for existing buildings: Literature review and future needs," *Automation in Construction*, vol. 38, pp. 109-127, 2014, doi: 10.1016/j.autcon.2013.10.023.

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[3] N. Abreu, A. Pinto, A. Matos, and M. Pires, "Procedural point cloud modelling in scan-to-BIM and scan-vs-BIM applications: A review," *ISPRS International Journal of Geo-Information*, vol. 12, no. 7, Art. no. 260, 2023, doi: 10.3390/ijgi12070260.

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